

# PG-S3: Physically-Grounded Selective State Space Models

## Research Overview

Modern AI systems increasingly need to simulate physical environments — predicting how a robot arm will move, how a fluid will flow, or how a dynamical system will evolve over long time horizons. These tasks are the domain of *world models*: learned simulators that roll out future states from a compact latent representation. The dominant approach uses Transformer architectures, but Transformers carry a fundamental computational burden — their self-attention mechanism scales as  $O(L^2)$  in both time and memory with respect to sequence length  $L$ . On consumer-grade, CPU-bound hardware, this makes long-horizon simulation prohibitively expensive.

Selective State Space Models (SSMs), most notably Mamba, recover  $O(L)$  complexity through a recurrent scan, making them attractive for deployment on constrained hardware. However, they carry a structural flaw when applied to physical systems: their standard Zero-Order Hold (ZOH) discretization treats the latent transition as a piecewise-constant step, with no mechanism to enforce the conservation laws that govern real physics. Specifically, ZOH does not constrain the transition Jacobian to be symplectic — it does not preserve the canonical two-form of Hamiltonian phase space. The consequence is artificial energy drift that compounds over long rollouts, steadily corrupting the physical fidelity of the simulation.

This paper introduces **PG-S3 (Physically-Grounded Selective State Space)**, a world-model architecture that closes this gap. It achieves  $O(L)$  computational complexity while rigorously preserving Hamiltonian geometric structure, and does so within the memory and compute envelope of an 8 GB consumer CPU.

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## Core Technical Contributions

**1. Symplectic Selective Dynamics.** The central innovation is replacing ZOH with a *Störmer–Verlet symplectic integrator* inside the SSM scan. The latent state is partitioned into generalised coordinates and momenta,  $h = [q, p]^T$ , encoding a Hamiltonian phase-space vector. The Störmer–Verlet update — a half-kick in momentum, a full drift in position, then another half-kick — is

provably symplectic: its Jacobian  $M$  satisfies  $MTJM = J$ , preserving phase-space volume exactly at every step. This means the integrator conserves a *shadow Hamiltonian*  $\tilde{H} = H + O(\Delta t^2)$  over exponentially long intervals, with energy error that oscillates within a bounded band rather than growing secularly. The formal proof invokes the composition of exact Hamiltonian flows, a result from geometric numerical integration theory.

**2. Gershgorin CFL Stability Bound.** For a latent transition matrix  $A$ , the Gershgorin Circle Theorem provides a conservative upper bound  $\rho_G$  on the spectral radius. Since Störmer-Verlet stability for a harmonic mode requires  $\omega \cdot \Delta t \leq 2$ , the paper derives the Latent CFL Condition:  $\Delta t = \min(\Delta_{\max}, 2\eta/\rho_G)$ , where  $\eta$  is a safety factor. This bound is computed branchlessly via AVX2 horizontal reduction, contracting the time step automatically during high-gradient perturbations without pipeline stalls — a hardware-aware solution that prevents numerical blow-up without any conditional branching overhead.

**3. Scale-Adaptive INT4 Quantization (SAQ).** To fit the model within an 8 GB hardware budget, projection weights are quantized to 4-bit integers. Rather than using a fixed quantization scale, SAQ adapts the scale  $\alpha(t)$  per 64-element tile by tracking the local dynamic range of each tile's weights. The theoretical justification links the quantization noise floor  $\sigma^2_q$  to the tail energy of a Kolmogorov turbulence spectrum  $E(k) \propto k^{-5/3}$ , ensuring INT4 precision does not catastrophically suppress the physically relevant spectral content.

**4. Neural-Physical Coupling Layer.** The encoder maps input tokens to Hamiltonian phase space through a sequence of three elementary symplectic shear matrices  $W = S_3 S_2 S_1$ , each in  $SL(2, \mathbb{R})$ . Since for one degree of freedom  $SL(2, \mathbb{R}) = Sp(2, \mathbb{R})$ , the composition satisfies  $WTJW = J$  by construction. For the full  $d$ -dimensional latent space, symplecticity is preserved via a block-diagonal product of  $d/2$  independent shear pairs. Dissipative environments are handled through an operator-split Rayleigh damping step — deliberately kept outside the conservative symplectic core — with the conformally symplectic update  $p \leftarrow \exp(-\gamma \Delta t) \odot p$  guaranteeing monotonic energy removal.

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## Empirical Validation

The framework is validated across five experimental phases on a single Intel Core i5 CPU (8 GB RAM, AVX2, macOS Ventura).

**Phase 01** benchmarks scan latency across sequence lengths from 128 to 16,384. Log-log regression confirms  $O(n)$  empirical scaling with  $R^2 = 0.996$ , and peak RSS remains well below 5 GB at all lengths.

**Phase 02** subjects the Störmer-Verlet kernel to a controlled harmonic-oscillator audit ( $\omega = 1$ ,  $\Delta t = 0.01$ , 1000 steps). The maximum energy-conserving error  $ECE_{\max} = 2.5000 \times 10^{-5}$  matches the theoretical shadow-Hamiltonian bound  $\omega^2 \Delta t^2 / 4$  to four significant figures. The secondary check  $ECE_{\text{rms}} \approx ECE_{\max} / \sqrt{2}$  confirms the sinusoidal identity expected for a bounded orbit — direct evidence of motion on the shadow energy surface rather than outward spiralling.

**Phase 03** validates the coupled system on 2D Navier-Stokes and rigid-body rollouts. A fourth-order vorticity stencil yields a mean-square-curl VPS of  $1.25 \times 10^{-9}$ , confirming physical fluid structure is preserved. Memory stays within the 5 GB ceiling.

**Phase 04** scales to 524,288 simultaneous trajectories in 512-dimensional phase space. The system sustains  $1.2 \times 10^6$  steps per second with a total fractional Hamiltonian drift of only  $1.26 \times 10^{-11} E_0$  across  $\sim 1.2 \times 10^9$  integration steps — eleven orders of magnitude below the energy scale. Peak worker RSS is 4,096.5 MB.

**Phase 05** applies topological reconstruction to the production attractor using delay embedding ( $\tau = 1$ ,  $m = 3$ ), the Rosenstein algorithm for Lyapunov exponent estimation, Poincaré sectioning, and correlation-dimension analysis. Results:  $\lambda_{\max} = 0.3578$  ( $R^2 = 0.9792$ ), predictability horizon  $L_{\text{crit}} = 2.79$  embedding steps, and correlation dimension  $D_2 = 2.12$ , consistent with a low-complexity dissipative strange attractor.

A direct ablation against forward Euler integration shows the symplectic advantage reaching  $268,400\times$  at  $10^4$  steps — a ratio that grows without bound as Euler diverges while PG-S3 remains bounded.

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## Conclusion

PG-S3 demonstrates that physical conservation laws and hardware-level efficiency are complementary rather than competing constraints. By grounding selective state space dynamics in Hamiltonian geometry, the architecture enables accurate, stable, long-horizon physical simulation at over one million steps per second on accessible consumer hardware — bridging the gap

between physics-informed neural networks and the linear-time efficiency demanded by real-world deployment.